

2-Mark Questions & Answers — Data Structures Module 6: Sorting & Searching Techniques

Q1. Differentiate between Linear Search and Binary Search.

- **Linear Search:** Unsorted/sorted arrays, sequential scan, $O(N)$ time.
 - **Binary Search:** Strictly sorted arrays, divide-and-conquer interval halving, $O(\log N)$ time.
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Q2. What is a Stable Sorting algorithm? Give examples.

A sorting algorithm is **Stable** if it preserves the relative order of duplicate elements with equal keys.

- **Stable Sorts:** Bubble Sort, Insertion Sort, Merge Sort.
 - **Unstable Sorts:** Selection Sort, Quick Sort, Heap Sort.
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Q3. Define Hashing and Collision.

- **Hashing:** Mapping a search key K to a table index $h(K)$ using a hash function.
 - **Collision:** When two distinct keys k_1 and k_2 produce the same hash index ($h(k_1) = h(k_2)$).
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Q4. What is Linear Probing vs Quadratic Probing?

- **Linear Probing:** Probes linearly $h(k, i) = (h'(k) + i) \bmod m$. Suffers from primary clustering.
 - **Quadratic Probing:** Probes quadratically $h(k, i) = (h'(k) + c_1 i + c_2 i^2) \bmod m$. Eliminates primary clustering.
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